Name – Meenakshi

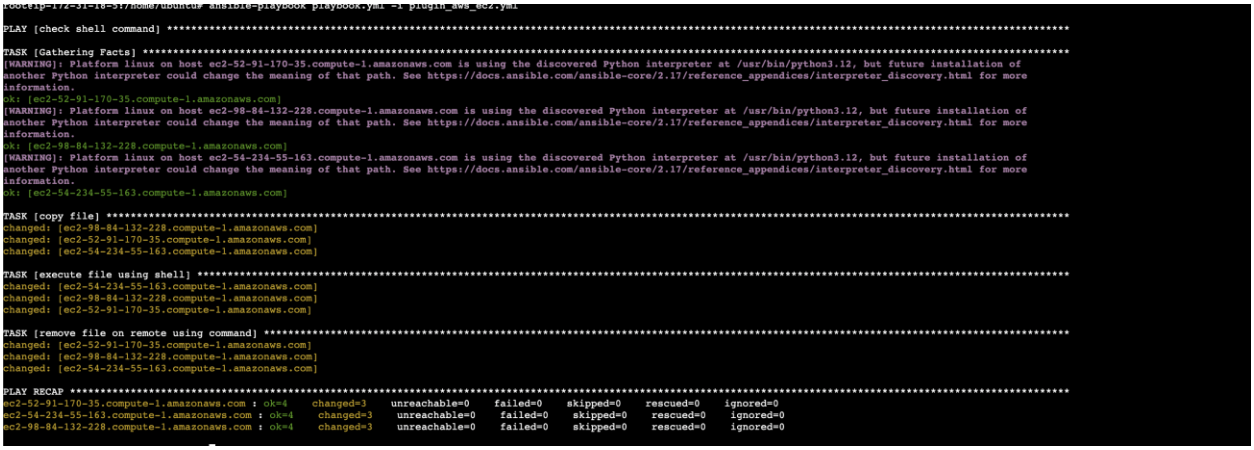
Batch - 7943

Tasks:

- Run Ansible ad-hoc commands to:
    - Execute the command in Bash, view the result, and check the difference between the shell and command modules
    - Check all the running processes on nodes
    - Copy files to multiple nodes concurrently
    - Add and delete users

- Create the ansible playbook to install docker on the Ubuntu machines
- Using loops, copy multiple files from the Ansible machine to the remote node
- Install MySQL on remote machines using Ansible modules
-

Execute the command in Bash, view the result, and check the difference between the shell and command modules



```
# Note that this file was always incomplete  and lagging changes to configurat
# for example, for 2.9: https://github.com/ansible/ansible/blob/stable-2.9/exa
[defaults]
host_key_checking=False
ansible_ssh_user=ubuntu
remote_user=ubuntu
private_key_file=/home/ubuntu/ansible.pem
```

```
  GNU nano 7.2                                                playbook.ym
---
- name: check shell command
  hosts: aws_ec2
  become: true
  tasks:
    - name: copy file
      copy:
        src: nginx.sh
        dest: /home/ubuntu/nginx.sh
        mode: 0755
    - name: execute file using shell
      ansible.builtin.shell: sudo sh nginx.sh
    - name: remove file on remote using command
      ansible.builtin.command:
        cmd: sudo rm /home/ubuntu/nginx.sh
```

```
plugin: amazon.aws.aws_ec2
regions:
  - us-east-1
keyed_groups:
  - key: tags
    prefix: tag
```

Check all running processes

```
TASK [Display running processes] ***********************************************************************************************************************************
ok: [ec2-54-234-55-163.compute-1.amazonaws.com] => {
    "process_list.stdout_lines": [
        "USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND",
        "root           1  0.0  1.3  22504 13532 ?        Ss   17:12   0:02 /sbin/init",
        "root           2  0.0  0.0      0     0 ?        S    17:12   0:00 [kthreadd]",
        "root           3  0.0  0.0      0     0 ?        S    17:12   0:00 [pool_workqueue_release]",
        "root           4  0.0  0.0      0     0 ?        I<   17:12   0:00 [kworker/R-rcu_g]",
        "root           5  0.0  0.0      0     0 ?        I<   17:12   0:00 [kworker/R-rcu_p]",
        "root           6  0.0  0.0      0     0 ?        I<   17:12   0:00 [kworker/R-slub_]",
        "root           7  0.0  0.0      0     0 ?        I<   17:12   0:00 [kworker/R-netns]",
        "root          10  0.0  0.0      0     0 ?        I<   17:12   0:00 [kworker/0:0H-events_highpri]",
        "root          11  0.0  0.0      0     0 ?        I    17:12   0:00 [kworker/u30:0-ext4-rsv-conversion]",
        "root          12  0.0  0.0      0     0 ?        I<   17:12   0:00 [kworker/R-mm_pe]",
        "root          13  0.0  0.0      0     0 ?        I    17:12   0:00 [rcu_tasks_rude_kthread]",
        "root          14  0.0  0.0      0     0 ?        I    17:12   0:00 [rcu_tasks_trace_kthread]",
        "root          15  0.0  0.0      0     0 ?        S    17:12   0:00 [ksoftirqd/0]",
        "root          16  0.0  0.0      0     0 ?        I    17:12   0:00 [rcu_sched]",
        "root          17  0.0  0.0      0     0 ?        S    17:12   0:00 [migration/0]",
        "root          18  0.0  0.0      0     0 ?        S    17:12   0:00 [idle_inject/0]",
        "root          19  0.0  0.0      0     0 ?        S    17:12   0:00 [cpuhp/0]",
        "root          20  0.0  0.0      0     0 ?        S    17:12   0:00 [kdevtmpfs]",
        "root          21  0.0  0.0      0     0 ?        I<   17:12   0:00 [kworker/R-inet_]",
        "root          23  0.0  0.0      0     0 ?        S    17:12   0:00 [kauditd]",
        "root          24  0.0  0.0      0     0 ?        S    17:12   0:00 [khungtaskd]",
        "root          25  0.0  0.0      0     0 ?        S    17:12   0:00 [oom_reaper]",
        "root          27  0.0  0.0      0     0 ?        I<   17:12   0:00 [kworker/R-write]",
        "root          28  0.0  0.0      0     0 ?        S    17:12   0:00 [kcompactd0]",
```

```yaml
    - name: Get list of running processes
      ansible.builtin.shell: ps aux
      register: process_list

    - name: Display running processes
      debug:
        var: process_list.stdout_lines
```

Create and delete

```
}

TASK [Add a user] **************************************************************************************************
ok: [ec2-52-91-170-35.compute-1.amazonaws.com]
ok: [ec2-54-234-55-163.compute-1.amazonaws.com]
ok: [ec2-98-84-132-228.compute-1.amazonaws.com]

TASK [Delete a user] ***********************************************************************************************
ok: [ec2-52-91-170-35.compute-1.amazonaws.com]
ok: [ec2-54-234-55-163.compute-1.amazonaws.com]
ok: [ec2-98-84-132-228.compute-1.amazonaws.com]
```

```yaml
    - name: Add a user
      ansible.builtin.user:
        name: meenakshi
        state: present  # Ensures the user is created
        shell: /bin/bash
        home: /home/meenakshi
        groups: sudo  # Add the user to the sudo group

    - name: Delete a user
      ansible.builtin.user:
        name: tempuser
        state: absent  # Ensures the user is removed
        remove: true  # Removes the user's home directory and mail spool
```

Install docker



```
- name: Install Docker on Ubuntu
  hosts: all  # Replace 'all' with your target group (e.g., 'web')
  become: true  # Required for elevated privileges
  tasks:
    - name: Update apt package index
      ansible.builtin.apt:
        update_cache: yes

    - name: Install required packages for Docker
      ansible.builtin.apt:
        name:
          - apt-transport-https
          - ca-certificates
          - curl
          - software-properties-common
        state: present

    - name: Add Docker's official GPG key
      ansible.builtin.command: curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o /usr/share/keyrings/docker-archive-keyring.gpg
      args:
        warn: false

    - name: Set up Docker repository
      ansible.builtin.apt_repository:
        repo: deb [arch=amd64 signed-by=/usr/share/keyrings/docker-archive-keyring.gpg] https://download.docker.com/linux/ubuntu {{ ansible_distribution_release }} stable
        state: present
        update_cache: yes

    - name: Install Docker Engine
      ansible.builtin.apt:
        name:
          - docker-ce
          - docker-ce-cli
          - containerd.io
        state: present

    - name: Start and enable Docker service
      ansible.builtin.service:
        name: docker
        state: started
        enabled: yes

    - name: Add the user to the docker group (optional)
```

- name: Install Docker on Ubuntu

  hosts: all # Replace 'all' with your target group (e.g., 'web')

  become: true # Required for elevated privileges

  tasks:

    - name: Update apt package index

      ansible.builtin.apt:

        update_cache: yes

    - name: Install required packages for Docker

      ansible.builtin.apt:

        name:

          - apt-transport-https

          - ca-certificates

          - curl

          - software-properties-common

        state: present

    - name: Add Docker's official GPG key

      ansible.builtin.command: curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o /usr/share/keyrings/docker-archive-keyring.gpg

      args:

```
    warn: false

  - name: Set up Docker repository
    ansible.builtin.apt_repository:
      repo: deb [arch=amd64 signed-by=/usr/share/keyrings/docker-archive-keyring.gpg] https://download.docker.com/linux/ubuntu {{ ansible_distribution_release }} stable
      state: present
      update_cache: yes

  - name: Install Docker Engine
    ansible.builtin.apt:
      name:
        - docker-ce
        - docker-ce-cli
        - containerd.io
      state: present

  - name: Start and enable Docker service
    ansible.builtin.service:
      name: docker
      state: started
      enabled: yes

  - name: Add the user to the docker group (optional)
    ansible.builtin.user:
      name: "{{ ansible_user }}"
```

groups: docker

append: yes

- **Using loops, copy multiple files from the Ansible machine to the remote node**
- **Install MySQL on remote machines using Ansible modules**

```yaml
- name: Copy files and install MySQL
  hosts: all  # Replace 'all' with your target group (e.g., 'db')
  become: true  # Required for elevated privileges
  vars:
    files_to_copy:
      - src: file1.txt
        dest: file1.txt
      - src: file2.txt
        dest: file2.txt
  tasks:
    - name: Copy multiple files to remote nodes
      ansible.builtin.copy:
        src: "{{ item.src }}"
        dest: "{{ item.dest }}"
        owner: root
        group: root
        mode: '0644'
      loop: "{{ files_to_copy }}"

    - name: Update apt package index
      ansible.builtin.apt:
        update_cache: yes

    - name: Install MySQL server
      ansible.builtin.apt:
        name: mysql-server
        state: present

    - name: Start and enable MySQL service
      ansible.builtin.service:
        name: mysql
        state: started
        enabled: yes

    - name: Secure MySQL installation (optional)
      ansible.builtin.shell: |
```

```yaml
        enabled: yes

    - name: Secure MySQL installation (optional)
      ansible.builtin.shell: |
        mysql -u root <<EOF
        ALTER USER 'root'@'localhost' IDENTIFIED WITH 'mysql_native_password' BY 'your_password';
        DELETE FROM mysql.user WHERE User='';
        DROP DATABASE IF EXISTS test;
        FLUSH PRIVILEGES;
        EOF
      args:
        warn: false
```